AEGIS RESEARCH INSTITUTE

DEEP TECHNICAL INTELLIGENCE REPORT

AI Interview Rooms:

Anatomy of Fraud, Biometric Defense,

and Advanced Gesture Intelligence

How the World's Leading Tech Companies Conduct AI-Powered Interviews — How Candidates Bypass Every Layer of Protection — And How Advanced Gesture + Eye Tracking Closes Every Gap

71%	Candidates admit to cheating
80%	Use LLMs on coding tests
25%	Profiles will be fake by 2028 (Gartner)
99.49%	CAST deepfake detection AUC
$50K+	Cost of a single bad hire

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1. EXECUTIVE SUMMARY & TL;DR INTELLIGENCE BRIEF

What This Report Covers

This is the most comprehensive technical intelligence document ever assembled on AI-powered interview room technology. It covers three parallel arms races simultaneously: (1) how the world's leading tech companies architect and deploy AI interview systems; (2) the complete technical anatomy of every known method candidates use to bypass these systems; and (3) the full scientific basis and implementation blueprint for advanced gesture recognition and eye tracking as the most effective countermeasures available today.

Key Intelligence Findings

71%	of job seekers admit to cheating during the hiring process (Resume Templates, 2025)
80%	of candidates use LLMs on coding tests despite explicit prohibition (Karat Research)
81%	of FAANG interviewers suspect AI cheating; 33% have caught someone (Interviewing.io)
55.54%	human deepfake detection accuracy — statistically indistinguishable from random chance
99.49%	AUC achieved by CAST CNN-Transformer on FaceForensics++ dataset
$5.3M	raised by Cluely (Interview Coder) — 'zero documented cases of detection'
3,000%	spike in deepfake fraud attempts in 2023 alone
97.5%	gaze tracking accuracy in controlled environments (MediaPipe Iris + Kalman Filter)
93.7%	F1-score for hand gesture fraud detection using MediaPipe (IJSRA Research)
25%	of candidate profiles projected to be entirely fake by 2028 (Gartner)

2. THE GLOBAL AI INTERVIEW ECOSYSTEM

2.1 Four Categories of AI Interview Platforms

The AI interview market has stratified into four architecturally distinct categories, each addressing different stages of the hiring pipeline with fundamentally different technical approaches. Understanding this taxonomy is essential for understanding where each category's security vulnerabilities lie.

CATEGORY	PLATFORMS	ARCHITECTURE	SECURITY EXPOSURE
Live AI Interviewers (Bidirectional)	Fabric, Talently.ai, Goodfit	Real-time NLP + STT + Adaptive Q-Gen (0.8s latency)	Highest — live stream manipulation window
Pre-Recorded Video (Async Scoring)	HireVue, Interviewer.AI, Spark Hire	Asynchronous CV analysis, transcript scoring, rubric match	Medium — deepfake video injection
Technical Assessment (Code Evaluation)	CodeSignal, HackerRank, HackerEarth, Codility	Live IDE, pair-programming, algorithmic auto-grading	High — invisible overlay tools (Cluely)
Interview Intelligence (Human Augmentation)	BrightHire, Metaview, Talview	Recording + NLP coaching, structured interview notes	Low — augments human interviewer judgment

2.2 How the World's Top Companies Run AI Interviews

Each major technology company has developed a distinct philosophy toward AI-assisted hiring, creating different attack surfaces and requiring different defensive strategies.

COMPANY / PLATFORM	AI INTERVIEW METHOD	UNIQUE TECH	ADOPTION SCALE
HireVue (Fortune 500 dominant)	Asynchronous video + game-based cognitive assessments	Removed facial analysis 2021 (EPIC/FTC complaint). Now: transcript + tone + word-choice scoring	30M+ interviews processed; ~50% of Fortune 100
Talview (Enterprise + Campus)	Live + async proctoring with 'Seven-Layer Framework'	Smart device detection, room scan, behavioral AI, 50+ languages, dual-camera	95% detection accuracy claimed; 8x higher than traditional
Mercer Mettl (Assessment + Interview)	Proctored assessments + 3-point authentication	26+ question formats, Bluetooth scanner, keystroke biometrics	Campus hiring at 100K+ concurrent scale
Goodfit (AI-Native, India)	Conversational voice AI — no fixed app required	Multilingual foundation models (14 Indic languages), 0.8s follow-up, dual-model auditing	Starting at Rs.100/interview; high-volume mid-market
CodeSignal (Technical Hiring)	Live IDE + pair programming + AI-powered scoring	FedRAMP certified; 40+ languages; sandboxed execution environment	FAANG and top-100 tech companies; US-dominant
Pymetrics / Harver (Neuroscience-Based)	12 gamified cognitive and emotional assessments	Neuroscience game battery; behavioral trait profiling; bias-aware ML matching	Unilever (cut interview time by 75%); global enterprise
Interviewer.AI (India, Mobile-First)	Sentiment analysis + competency rubric scoring	Optimized for 4G networks; mobile-first design	Rs.1,000-2,900 per interview; early-stage companies
Fabric (Detection-First)	Live AI conversation + 20+ forensic signals simultaneously	Adaptive questioning breaks AI coherence; behavioral consistency modeling	85% cheating detection; 100+ signal analysis

2.3 Platform Pricing and Market Dynamics (India Focus)

PLATFORM	INDIA PRICING	TARGET SEGMENT	PROCTORING LEVEL
Interviewer.AI	Rs. 1,000 - 2,900 / interview	Early-stage	Basic (face + audio)
Talview	Rs. 1,650 - 4,150 / interview	Banking + Enterprise	Full 7-layer
Spark Hire	Rs. 4,150 - 12,450 / month	Startups + Agencies	Medium
Goodfit	From Rs. 100 / interview	High-volume mid-market	Behavioral + DOM
Mercer Mettl	Custom enterprise	Campus + Global events	Full + Bluetooth scan
OmniHire	Custom service model	Pan-India remote	AI shortlisting + human
HireVue	Rs. 29L - 1.2Cr / year	Fortune 500 India ops	Advanced (no face emotion)
ClearRound.ai	Rs. 199 / week	Individual candidates	N/A (cheating tool)

3. THE COMPLETE THREAT MODEL — HOW CANDIDATES CHEAT

3.1 Scale of the Problem

71%	admit cheating (Resume Templates)
80%	LLMs on code tests (Karat)
81%	FAANG suspect cheating
3000%	deepfake fraud spike 2023
6.5%	of all fraud is deepfake

The economic calculus is brutal: cheating tools cost $20-50/month while the positions they target pay $150,000+ annually. A single bad hire costs organizations $50,000+ in wasted salary, team productivity loss, and re-hiring expenses. The risk-reward ratio overwhelmingly favors the cheater under current detection regimes.

3.2 Layer 1 — Invisible Overlay Tools: The Cluely Architecture

Cluely (formerly Interview Coder) represents the most technically sophisticated category of cheating. Created by Columbia student Chungin 'Roy' Lee — expelled for using it to obtain an Amazon internship — the tool has raised $5.3 million and claims zero documented detections.

TECHNICAL MECHANISM	HOW IT WORKS	WHY CURRENT SYSTEMS MISS IT
Stateless Overlay (Core Exploit)	Renders at the OS compositor layer below the capture level. Operates as a transparent, click-through window.	Screen capture APIs see only the desktop, not the overlay. The overlay is invisible to screen sharing.
Background Daemon (Stealth Operation)	Runs as a background process invisible in Activity Monitor, Dock, and Taskbar. No visible windows or icons.	Cluely uses obfuscated
OCR Question Capture (Input Method)	Silently screenshots the screen at optimized intervals. Uses local OCR to extract question text.	No suspicious network traffic. Never types into the interview platform.
Local LLM / RAG (Response Generation)	Routes captured questions to a local RAG system or cloud LLM (GPT-4, Claude). Response appears as a floating text box.	Response delivered via keyboard injection.
Cross-Platform Support (Universal Reach)	Works on Zoom, Teams, Google Meet, Amazon Chime, Webex, HackerRank, CodeSignal, CoderPad, and more.	Single codebase. OS-level, not app-level.

FinalRound AI takes a different approach — running as a browser tab that monitors audio via Web Speech API, sends to an LLM backend, and displays responses as DOM overlays. Detection requires analyzing WebSocket connection patterns, DOM mutation rates, and browser extension manifests.

3.3 Layer 2 — Deepfake Video and Voice Cloning

The deepfake threat operates on three tiers of escalating technical complexity, each requiring a different countermeasure:

TIER	TECHNIQUE	TOOLS	DETECTION DIFFICULTY
Tier 1 Basic	Real-time face-swap via virtual camera routing. Neu	DeepFaceLive, OBS Virtual Cam, Wav	Dynamic PAD/PAI fails, but A
Tier 2 Advanced	Voice cloning + lip-sync. Audio synthesis clones voi	ElevenLabs, RVC, Wav2Lip	samples, pressing - human detection at 55.54%
Tier 3 State-Level	Full synthetic identity: non-existent photorealistic fa	Custom GAN pipelines, KVM switches	Digital Injection (DVI)

3.4 Layer 3 — Proxy Networks, KVM Switches, and Lip-Sync

Proxy interview rings operate as organized syndicates, particularly prevalent in high-volume IT hubs. A technically proficient proxy attends on behalf of the applicant. In advanced 'lip-sync proxy' setups, the actual applicant is on camera and silently moves their mouth while the hidden proxy supplies spoken answers via hidden earpiece. KVM (Keyboard-Video-Mouse) switches allow overseas facilitators to remotely control domestic laptops — a hardware-level attack that bypasses all software-based remote desktop detection.

3.5 Layers 4-6 — Hardware, VM, and Wearable Exploits

EXPLOIT METHOD	TECHNICAL MECHANISM	CURRENT DETECTION STATUS
Virtual Machine Sandboxing	Testing environment installed inside a VM guest OS. Pr	Partially detected. PEBuilder, combined with deep system process
Bluetooth Earpiece	Real-time audio from external expert transmitted via mic	Two-Factor Bluetooth scanner. 97% detection rate during live
AR Glasses	Project digital text directly into user's field of vision. Appe	No standard eye-based detection. Requires physical inspection
Smartwatch (Wearable)	Displays AI-generated answers on wrist. Kinematic sig	Detectable via head-pose tracking (YOLO-Pose)
Secondary Phone (Bezel Placement)	Phone placed directly against laptop bezel, minimizing	Detectable via ambient light sensor. Displays AI chatbot responses

4. HOW EACH CHEATING PROTOCOL IS BYPASSED — TECHNICAL ANATOMY

This section provides a complete technical mapping of how every layer of current proctoring is bypassed, and what the residual detection signal is in each case.

PROCTORING CONTROL	BYPASS METHOD	RESIDUAL DETECTION SIGNAL
Browser Full-Screen Lockdown	Run proctoring inside VM guest; host OS unrestricted.	DOM-level session replay blocks impossible (focus loss < 10 ms)
Screen Recording / Share Detection	Cluely renders below WebRTC capture layer. FinalRound AI uses	No direct detection for dual-machine overlay
Face Liveness (Blink/Nod)	GoldPickaxe trojan harvests real liveness footage (nod, blink)	Pupil-to-light reflex lag > 4 frames; sharp contrast
Head Pose Threshold (>45 deg)	'Head-forward, eyes-away' exploit: skull stationary, pupils shift	Gaze-to-pose discrepancy analysis
Audio Monitoring (Secondary Voice)	Bluetooth micro-earpiece: no ambient audio generated.	Expert speaks into a phone mic; candidate's MAR (Mouth A
Object Detection (Phone/Notes)	Phone placed in bezel blind spot below webcam field-of-view.	Gaze vector fixated outside frame (reading from phone).
Keystroke Biometrics (Typing Pattern)	Clipboard injection: Cluely injects code at OS level without	DOM-level keystroke telemetry: zero keydown events
Identity Verification (Face Match at Login)	Deepfake overlay at login only, then real candidate appears	Continuous face embedding cosine similarity drop

5. EYE TRACKING — THE SCIENCE OF GAZE-BASED DETECTION

5.1 MediaPipe Iris + Kalman Filter — The Production Architecture

The dominant eye-tracking pipeline for browser-based proctoring combines Google MediaPipe FaceMesh (468 3D facial landmarks) with MediaPipe Iris (iris, pupil, and eye contour) and a Kalman Filter for temporal smoothing. This pipeline runs entirely on the candidate's device via WebAssembly/WebGPU, achieving sub-30ms latency without any server round-trip.

COMPONENT	SPECIFICATION	PROCTORING APPLICATION
MediaPipe FaceMesh	468 3D facial landmarks; 30ms latency on CPU; runs via WebAssembly/WebGPU in browser	Provides dense eye region mesh for gaze and pose analysis;
MediaPipe Iris Model	Iris, pupil, and eye contour landmarks; trained on ~50,000 images; sub-pixel accuracy	Precise pupil center tracking enables gaze vector estimation via camera geometry
Kalman Filter (2D State)	Models gaze vector position and velocity; predicts next state; corrects with measurement	Eliminates jitter from natural micro-saccades; smooths trajectory for pattern
L2CS-Net (Alternative)	Joint classification-regression; outputs pitch and yaw as independent angles; ~50ms latency	Higher accuracy (3.92° MAE) for head pose variation; suitable for
Depth Estimation (Pinhole Model)	d = (real_iris_width * focal_length) / iris_pixel_width; <10% relative error at 40–80cm	Normalizes gaze vectors across candidates who shift position; corrects for

5.2 The Deviation Angle Formula and Proctoring Application

The core gaze detection algorithm computes the deviation angle between the eye gaze vector and the camera center axis:

```
theta = arccos( (G_vec . C_vec) / (|G_vec| * |C_vec|) )
```

Where G_vec is the gaze direction vector derived from iris landmarks and C_vec is the camera axis vector. A deviation exceeding 25 degrees triggers a cheating alert, but the system requires this threshold to be exceeded for 3+ consecutive frames to reduce false positives from natural eye movements. The Kalman-filtered hybrid achieves 97.5% accuracy in controlled environments, dropping to 92.3% in low-light and 85.2% under head motion — still above the 80% operational threshold.

5.3 Fixation, Saccade Analysis, and Whitespace Fixation Detection

The most powerful application of gaze tracking is not angle thresholding but behavioral pattern classification. Eight major gaze features have been identified as fraud indicators in systematic literature review:

GAZE FEATURE	NORMAL COGNITIVE PATTERN	CHEATING PATTERN	DETECTION METHOD
Fixation (most common feature, 19/37 studies)	100-400ms still periods during active thinking; distributed across screen	High-duration fixations on single region (reading answer); Fixation clustering in whitespace (looking at notes)	Spatial clustering; duration analysis
Saccades (rapid eye movements)	Natural ballistic jumps between fixation points; velocity-dependent	Mechanical, uniform saccades; Saccades directed toward edges (typing)	Velocity profile analysis; trajectory
Pupil Size (2nd most common, 10/37 studies)	Varies with cognitive load, lighting, emotion; natural baseline	Dilation during suspicious moments; constriction when looking at notes	Baseline comparison; response latency
Blink Rate	Natural: 15-20 blinks/minute; brief closure 150-400ms duration	Reduced blinking during intense reading; Eye blink overlay tracking	Blink rate statistical deviation
Gaze Entropy (randomness)	High entropy during genuine thinking (varied, non-repetitive)	Low entropy (eyes drift to same area repeatedly); Sustained gaze pattern	Shannon entropy over 10-second window
Off-Screen Gaze (duration)	Brief natural glances 0-2 seconds; return to screen	Sustained off-screen fixation >3 seconds; Repeated off-screen glances	Duration threshold; frequency analysis

GAZE FEATURE	NORMAL COGNITIVE PATTERN	CHEATING PATTERN	DETECTION METHOD
Whitespace Fixation	Candidate should have near-zero fixation on whitespace (looking at camera/thinking)	Cluttered whitespace (looking at notes)	Heatmap analysis
Gaze-Response Temporal Correlation	Eye movements precede or coincide with speech onset	Eyes move after speech begins (reading)	Temporal cross-correlation

5.4 Pupillary Light Reflex (PLR) and rPPG — Active Liveness

The two most powerful liveness signals available from a standard RGB webcam are the Pupillary Light Reflex (PLR) and Remote Photoplethysmography (rPPG). Neither can be reliably replicated by photos, pre-recorded videos, or current deepfake generators.

TECHNIQUE	MECHANISM	DEFEATS	ACCURACY	IMPLEMENTATION
Pupillary Light Reflex (PLR)	System rapidly cycles screen brightness. Live pupil constricts (bright) and dilates (dark) in 200-500ms	Photo attacks (TPR 99.6% at 1.0% FPR); 3D mask attacks (TPR 99.0% at 1.0% FPR)	99.0-99.4% TPR @ 10% FPR	Analyze pupil response to brightness changes
Remote Photo-plethysmography (rPPG)	Analyzes minute cyclical variations in skin color caused by blood flow (heartbeat)	Sophisticated deepfakes (no physiological signal); static photos; 3D masks	Feasible detection; approaches	Extract rPPG signal from face region, check for periodic pulse
PLR + rPPG Combined Check (NOVEL APPROACH)	Single 10-second active challenge: screen flashes to elicit PLR while rPPG is extracted	All photo/video attacks; most 3D attacks; real-time deepfakes	Theoretical: >99.5% combined (no published benchmark yet)	This is the core innovation of this project

5 (CONT.) — GAZE SIGNATURES + STATISTICAL BASELINE MODELING

5.5 Gaze Pattern Signatures of Cheating Behavior

BEHAVIORAL SIGNATURE	INDICATES	DETECTION CONFIDENCE	TECHNICAL BASIS
Horizontal L-to-R scanning with quick snap-back	Reading from hidden script or invisible overlay	High (94.4% F1 — IJSRA)	Saccade velocity pattern + trajectory asymmetry ratio
Repeated off-screen fixation >3 seconds in one direction	Hidden phone / second device / notes	High	Gaze entropy analysis + duration threshold + spatial clustering
Consistent downward gaze during response	Notes on desk or phone below camera	Medium-High	Pitch angle negative deviation from neutral (FloCareer)
Whitespace fixation concentration	Cluely overlay or bezel-mounted phone (novel approach)	High (novel approach)	AI-detectable Cluely signature; gaze-vs-whitespace density KDE
Gaze aversion during flash (PLR test)	Photo / video / deepfake — no genuine pupil response	Very High (99.39% AUC)	Screen brightness change + pupil constriction measurement
Mechanical fixation + fluent speech simultaneously	Reading AI-generated response while speaking — not conversing	High	Temporal gaze-speech correlation; reading pattern during audio ac
Consistent 3-4 second pause before EVERY answer	'Flatline response timing' — AI processing delay	Medium-High	Response latency distribution analysis; variance too low = AI
Top-right quadrant fixation concentration	Hidden notes / second screen in top-right physical location	High	Screen quadrant gaze density mapping; top-right vs. content ratio

5.6 Statistical Baseline Deviation — The KS-Test Implementation

The critical upgrade from commercial systems: replace raw threshold-based gaze flags (if angle > 25° for 3 frames = flag) with per-candidate statistical baseline modeling using the Kolmogorov-Smirnov test. This is documented in literature but implemented in no open-source proctoring pipeline.

Implementation Steps:

1. CALIBRATION PHASE (first 90 seconds): Collect candidate's natural gaze distribution. Build baseline gaze point cloud. Record natural fixation patterns, entropy levels, saccade velocities.
2. CONTINUOUS MONITORING: Collect rolling 10-second gaze windows throughout interview.
3. KS-TEST APPLICATION: Apply two-sample Kolmogorov-Smirnov test comparing active window distribution against personal baseline. Flag ONLY when distribution shift is statistically significant: $p < 0.05$.
4. CONTEXTUAL SUPPRESSION: If candidate is responding verbally (audio activity detected), suppress gaze flags for natural cognitive drift. Flag only when gaze deviation coincides with sustained silence OR with AI-phrasing detected in concurrent speech analysis.
5. FUSION: Combine KS-test flag with other channels. Require 3+ concurrent channel deviations before raising integrity event.

Why this matters: A candidate with naturally wandering eyes will never be flagged because their active distribution matches their personal baseline. A candidate reading from an overlay will show a statistically significant shift to low-entropy, directionally concentrated gaze — regardless of where their eyes naturally wander when thinking.

6. ADVANCED GESTURE RECOGNITION — KINEMATIC SURVEILLANCE

6.1 BlazePose, MediaPipe, and OpenPose — Framework Comparison

FRAMEWORK	LANDMARKS	PARAMETERS	FPS (MOBILE)	KEY INNOVATION	PROCTORING USE
BlazePose Lite (MediaPipe)	33 body	1.3M	31 FPS (Pixel 2)	Two-stage detect-then-track + ROI refinement	Real-time skeletal mapping in browser via WebGL/Wasm
BlazePose Full (MediaPipe)	33 body	3.5M	30+ FPS	Heatmap + regression dual-head architecture	Higher accuracy; detects subtle arm and wrist movements
OpenPose	17-25 body	25.1M	23.7 FPS	Bottom-up PAF: identifies all body parts then groups	Multi-person: detects secondary individual in frame (crowded scenes)
YOLOv8-Pose (YOLO-Pose)	17 body	Custom	Real-time	Object detection + pose estimation in single pass	Simultaneous object (phone/notes) + pose detection
MediaPipe Hands	21/hand	Lightweight	30+ FPS	21 3D landmarks per hand including finger joints	Hand-to-ear detection; off-camera reach; burst typing patterns

6.2 33-Keypoint Skeletal Mapping — What Each Keypoint Catches

BlazePose's 33-landmark schema provides superior articulation coverage compared to OpenPose's 17-point COCO format. The additional landmarks (knuckles, heels, foot index) are not relevant for interview proctoring, but the upper-body landmarks provide the following fraud signals:

KEYPOINT GROUP	LANDMARK IDs	FRAUD SIGNAL DETECTED
Face landmarks (11, 12, 0-10)	Nose, eyes, ears, mouth corners	Head pose (yaw/pitch/roll); gaze direction cross-reference; lip movement for MAR calculation
Shoulders (11, 12)	Left and right shoulder	Asymmetrical shoulder drop = off-camera mobile typing biomechanics; forward lean = reading from hidde
Elbows (13, 14)	Left and right elbow	Arm trajectory analysis; motion toward ear (earpiece interaction); arm extended off-camera = reaching for
Wrists (15, 16)	Left and right wrist	Wrist rotation + downward head pitch = smartwatch interaction; wrist disappearing below frame = phone/k
Hands / Knuckles (17-22)	Pinky, index, thumb tips + palms	21-landmark MediaPipe Hand model activated for precision; hand-to-ear proximity detection; covering mo
Upper Arm Velocity	Computed from shoulder-elbow vector	Sudden rapid movement toward off-camera zone = fetching hidden device; kinematic signature distin

6.3 Specific Gesture Fraud Detection — Technical Implementation

GESTURE / ACTION	KINEMATIC SIGNATURE	DETECTION LOGIC	FALSE POSITIVE RISK
Hand-to-Ear (Earpiece)	Hand landmark trajectory: rapid upward movement toward ear region (landmark 7 or 8)	Hand detected in ear region for >0.8 seconds	Adjusting hair, scratching ear
Off-Camera Reach (Hidden Device)	Wrist landmark moves beyond frame_width * 0.85 threshold	Arm extension toward off-screen area	Reaching for water, gesturing
Burst Typing Pattern (Code Injection)	No keystroke events in DOM replay but large code block appears	Code block >50 lines appears within 1 second	Pasting from own notes
Smartwatch Interaction	Prolonged downward head pitch (>20 degrees) coupled with specific wrist motion	Head pitch + wrist motion sequence	Looking down to think
Multi-Person Detection	Second set of 33 landmarks detected in frame	MediaPipe Pose processes multiple persons	Person in background
Covering Mouth (Signal to Accomplice)	Both hands converge toward lower face region	Face MAR (Mouth Aspect Ratio) change + hand proximity	Yawning, coughing

6.5 Why Gesture + Gaze Fusion Overcomes Individual Limitations

DETECTION LAYER	ALONE (LIMITATIONS)	FUSED WITH OTHER LAYERS (RESULT)
Eye Tracking Only	False positives: natural cognitive drift, thinking pauses, looking at notes	Gaze + Hand: confirms what candidate is looking at (AND logic)
Gesture Recognition Only	Cannot confirm what candidate is looking at. Shoulder shrug = ambiguous	Gesture + Gaze: 93.7% F1 score, false positive rate drops to near-zero
Audio Analysis Only	Cannot isolate earpiece whisper from background noise	Audio + Gesture: earpiece whisper + hand-to-ear = definitive
Keystroke Dynamics Only	Cannot detect Cluely (no keystrokes simulated). Can detect DOM injection but not overlay	Keystroke + DOM: injection + paste = 99% accuracy for events
Content / NLP Only	LLMs can generate human-sounding answers. Vocabulary-based detection is unreliable	Content + Gaze: detectable vocabulary + off-screen gaze = near-definitive

7. DEEPFAKE DETECTION — CNN-TRANSFORMER ARCHITECTURES

7.1 CAST: Cross-Attentive Spatio-Temporal Architecture

CAST represents the current state-of-the-art in deepfake video detection. Its core innovation is a unified cross-attention mechanism enabling temporal features to dynamically attend to relevant spatial regions, enhancing detection of time-evolving artifacts like flickering eyes, warped lips, and inconsistent facial textures.

MODEL	CNN BACKBONE	AUC (FF++)	AUC (Celeb-DF)	AUC (DFD)	KEY MECHANISM
CAST-B5	EfficientNet-B5	99.49%	74.92%	92.07%	Cross-attention fusion
CAST-B0	EfficientNet-B0	98.98%	76.98%	93.31%	Better generalization
D2-Fusion	EfficientNet	96.87%	79.40%	91.90%	Dual-branch fusion
MAT	Swin Transformer	97.60%	67.40%	N/A	Mixed attention transformer
FaceGraphNet	ResNet+GNN+LSTM	N/A	0.96	0.96	Graph neural network hybrid
TimeSformer	Divided Attention	Sota K400	N/A	N/A	Space-time divided attention; 96 frames

CRITICAL LIMITATION: CAST's cross-dataset generalization drops from 99.49% on known datasets to 74.92% on unseen datasets (Celeb-DF). This 25-point accuracy gap is the active research frontier and the reason challenge-response approaches are being developed as primary alternatives.

7.2 Audio-Visual Lip-Sync Detection

Genuine speech produces deterministic biological correlations between acoustic frequencies and facial muscle movements. Synthetic media uses separate generation pipelines for audio and video, creating imperceptible synchronization failures detectable by AI.

METHOD	MECHANISM	DATASET / ACCURACY
Mouth Aspect Ratio (MAR)	MAR = geometric relationship between labial landmarks (dlib points 60, 62, 64, 66, 68). Tracks mouth openness; detects unnatural MAR values	AVLips (34,272 samples); 95.3% accuracy (LipFD)
SAVe (Self-Supervised AV)	Dual-headed model: processes visual lip sequence + audio spectrogram. Cross-modality alignment detection, real-time capable	Cross-dataset generalization; 2 alignments.
ViViT (Video Vision Transformer)	Whole-face temporal features: irregular blink rates, unnatural skin smoothing, hair/pixel inconsistencies	State-of-the-art video classification; applied to deepfake spatiotempor
Voice Style Detection	Monitors AI voice clone artifacts: missing acoustic micro-hesitations, robotic pitch, unnatural waveforms	Spectral analysis for 'microp

7.3 Challenge-Response: The Gotcha System (UC Berkeley)

Instead of passively detecting deepfake artifacts, Gotcha makes deepfake generation fail in real-time. Each challenge component exploits a deepfake pipeline vulnerability:

CHALLENGE TYPE	HOW IT BREAKS DEEPFAKES	DATA COLLECTED
Rapid head turns (extreme angles)	Face-swap pipelines have optimal operating angles; extreme rotation causes seamless 3D alignment failure and edge artifacts	47 real users, 58,247 real + fake videos; consistent v
Hand-over-face (partial occlusion)	Occlusion handling is the weakest point in real-time deepfake pipelines. Hand edges create detectable artifacts at occlusion boundaries	Spatial artifacts are detectable.
Lighting changes (lights on/off)	Deepfake renderers are trained on specific lighting distributions. Sudden lighting changes cause skin tone inconsistencies and shad	Shadow analysis

CHALLENGE TYPE	HOW IT BREAKS DEEPFAKES	DATA COLLECTED
Simultaneous speech + movement	Most deepfake pipelines process audio and video separately. Simultaneous complex multi-modal action causes synchronization fai	Same data as above

8. THE FIVE-LAYER ANTI-CHEAT DEFENSE ARCHITECTURE

Complete Defense-in-Depth Framework

LAYER	NAME	TECHNOLOGIES	CATCHES	MISSES / LIMITATIONS
Layer 1	Environmental Controls	Full-screen API, Page Visibility API, JavaScript event listeners	Tab switching, copy/paste, context menu, DevTools detection	Determined attackers can bypass (e.g., VM, phone camera)
Layer 2	Identity & Liveness	MediaPipe FaceMesh + Iris, PLR screen cycling, rPPG (remote photoplethysmography)	Photo attacks (99.9% TPR), static deepfakes (98% TPR), face-swap deepfakes (85% TPR)	Real-time deepfakes (CANDOR), high-quality video replay
Layer 3	Behavioral Monitoring	Gaze estimation (L2CS-Net), KS-test statistical analysis	Scripted reading (gaze pattern), coordinated multi-candidate fraud	Skilled actors can mimic natural gaze patterns
Layer 4	Audio Forensics	Speaker verification (GMM-UBM, i-vectors), voice activity detection	Ventriloquism, voice cloning, off-camera coaching	Whispered coaching, high-quality voice conversion
Layer 5	Human Oversight	Scrubable video timeline, Ghost View (CV overlay), Quick Jump to Layer 1–4 events	Edge cases, novel attack vectors, contextual judgment	Requires trained reviewer, time-consuming

System Performance Benchmarks

METRIC	TARGET	FALLBACK THRESHOLD	WHAT HAPPENS AT THRESHOLD
CV Inference Latency	<200ms (client-side)	200-500ms	Warn user; close background apps
Frame Rate	30 FPS minimum	<15 FPS	Alert user; switch to reduced-strictness mode
Telemetry Bandwidth	<50 KB/s per session	>100 KB/s	Switch to MessagePack compression
System Uptime	99.99% SLA	<99.9%	Kubernetes auto-scaling; multi-region failover
False Positive Rate	<2%	>2%	Retrain CV models; adjust KS-test p-value threshold
Deepfake Detection	>98% on known attacks	<90%	Escalate to challenge-response + human review
Data Retention	90-day auto-purge	Manual override only	AWS Lambda scheduled deletion

9. BEHAVIORAL BIOMETRICS — THE COGNITIVE FINGERPRINT

The most resilient anti-cheating defense is behavioral consistency modeling: building a comprehensive profile of the candidate across the entire interview and flagging statistically improbable patterns. This cannot be evaded because it requires the cheater to maintain realistic behavior across ALL dimensions simultaneously for the full interview duration.

BIOMETRIC CHANNEL	FEATURES MEASURED	FRAUD INDICATOR	ACCURACY
Keystroke Dynamics	Dwell time (key held duration), flight time (gap between keystrokes), typing speed, error rate, correction patterns	Typing speed >180 WPM with near-zero error rate; uniform inter-keystroke timing (machine-like); sudden change in rhythm mid-answer (pasting)	>99% (didit.me); 0.1% FAR, 1.0% FRR (Computers in Human Behavior); ~95% (TypingDNA)
Mouse Dynamics	Cursor speed, movement fluidity (Bezier curve analysis), scroll patterns, click pressure, idle movement	Unnaturally straight cursor paths; absence of micro-movements; clicks at regular intervals	~85% (TypingDNA)
Response Latency Distribution	Time from question end to response start, variance across questions, correlation with question complexity	Flat latency across all questions (reading from second screen); latency not correlated with question difficulty	82% (Facebook research)
Content Coherence (NLP)	Vocabulary complexity score per answer, topic consistency, grammatical structure, filler word patterns	Vocabulary jumps from 40th to 95th percentile between questions; AI-typical phrasing; perfect grammar with no filler words	89% (Frontiers in Psychology)
Application Navigation Flow	Window focus events, DOM mutation rate, clipboard interactions, tab switching, mouse exit from content area	WebSocket connection to external AI service; clipboard activity immediately before long answers; frequent tab switching	98% (Kovalchuk et al.)
CNN-LSTM-Attention (Session-Long Auth)	Combines keystroke + mouse + scroll into temporal sequences; CNN for local features, LSTM for temporal dependencies, Attention for important moments	Behavioral drift from established baseline; impersonation attempt detected by mismatch with enrolled profile	99.85% accuracy; <0.5% EER (Univ. Twente)

10. WHAT EXISTING SYSTEMS MISS — THE 5 UNSOLVED PROBLEMS

This section identifies the five critical gaps in current commercial and academic proctoring systems. These are the areas where the arms race is actively being lost — and where novel research contributions can establish genuine advancement over the state of the art.

GAP 1 — Compositor-Level Overlay Detection

PROBLEM: Cluely and similar tools render below the WebRTC capture layer at the OS compositor level. No current webcam-based, browser-based, or screen-capture-based system can detect these overlays directly. The only indirect signal is whitespace fixation — candidates using overlays fixate on UI whitespace where no content exists. This signal has never been implemented in a published proctoring pipeline.

SOLUTION: Whitespace fixation detection via UI content zone mapping + gaze density KDE analysis. Candidate's eyes should never concentrate on blank margins. This is the ONLY webcam-detectable Cluely signal available today.

GAP 2 — Per-Candidate Statistical Gaze Modeling vs. Universal Thresholds

PROBLEM: Every commercial system uses universal thresholds: angle > 25 degrees for 3 frames = flag. This produces high false-positive rates for candidates with naturally wandering eyes, medical conditions, or cultural differences in eye contact norms. The result is either high false positives or under-sensitivity.

SOLUTION: KS-test based per-candidate baseline deviation. Build candidate's gaze distribution in first 90 seconds. Flag only when active window distribution significantly deviates from personal baseline ($p < 0.05$). Zero published implementations exist.

GAP 3 — Combined PLR + rPPG Single Liveness Check

PROBLEM: Commercial systems use PLR OR rPPG, never both in a single integrated check. CAST achieves 99.49% AUC on known deepfake datasets but drops 25 points on unseen data. The fundamental problem is that detection systems are reactive — they recognize known deepfake styles. Physiological signals (PLR + rPPG) are proactive — they detect the absence of biology.

SOLUTION: 10-second combined check: screen flash triggers PLR while continuous rPPG monitors heartbeat. Both signals must be present and physiologically consistent. No published paper demonstrates this combination in a single RGB webcam check.

GAP 4 — Cross-Dataset Deepfake Generalization

PROBLEM: CAST drops from 99.49% (FF++) to 74.92% (Celeb-DF). The generalization gap means that new deepfake styles — particularly those using novel backbone architectures or trained on different data distributions — evade current detectors entirely until training data catches up. This is a fundamental limitation of pixel-artifact detection approaches.

SOLUTION: Challenge-response (Gotcha system) is the most promising direction: making deepfake generation fail rather than detecting its output. FaceGraphNet's geometric constraint approach (facial geometry proportions that deepfakes violate) shows better generalization than pixel-based methods.

GAP 5 — Five-Channel Behavioral Consistency Fusion

PROBLEM: Individual channels (gaze, gesture, audio, keystroke, content) each have 15-25% false negative rates when used alone. Commercial systems stack independent checks without statistical fusion — a candidate just needs to evade one channel. The fusion layer that requires 3+ simultaneous channel deviations with statistical significance is described in academic literature but not implemented in any open-source pipeline.

SOLUTION:	Late fusion architecture: each modality produces independent embeddings, concatenated and fed to final MLP classifier. Fusion layer sets N=3 minimum concurrent channel deviation threshold before raising integrity event. Makes evasion exponentially harder: cheater must simultaneously maintain believable gaze, gesture, audio, keystroke, and content patterns.

11. THE AEGIS FRAMEWORK — NEXT-GENERATION BLUEPRINT

11.1 Full System Architecture — Edge-Cloud Hybrid

COMPONENT	TECHNOLOGY	FUNCTION	DATA POLICY
Edge: CV Engine (Client Device)	TensorFlow.js / ONNX Runtime Web via WebAssembly + WebGPU; MediaPipe Face Mesh, Iris, Pose, Hands models	Real-time facial landmark detection, gaze vector estimation, head pose estimation, hand tracking, object detection — all on device. Only	Raw video never leaves device. Only
Edge: PLR + rPPG Liveness Module	JavaScript screen brightness API + FaceMesh forehead ROI pixel analysis	Controlled luminance flashes; measures pupillary light reflex latency and amplitude; extracts rPPG heart rate signal; fuses into liveness	Biometric signals processed and discarded locally; only
Edge: Whitespace Fixation Detector	UI layout coordinate extraction + gaze vector intersection	Continuously tracks if gaze is directed at empty whitespace regions (where	Gaze coordinates relative to viewport only; no
Transport Layer	WebSocket (telemetry) + WebRTC (media stream)	Secure, low-latency, bidirectional channel for telemetry events, heartbeats, and optional video	TLS 1.3 encryption; no persistent
Cloud: Message Ingestion	Apache Kafka cluster; auto-scaling consumer groups	Ingests high-frequency telemetry from thousands of concurrent sessions; decouples producers from	Kafka topic retention: 7 days;
Cloud: Analytics Microservices	FastAPI Python services; KS-test statistical engine;	Per-candidate baseline building; KS-test analysis; cross-channel fusion;	Derived features only; no raw
Storage: Relational (PostgreSQL)	PostgreSQL with PII encryption at application layer	User profiles, session metadata, hashed embeddings (not raw biometrics)	AES-256 encryption; cryptographic hash only.
Storage: Time-Series (TimescaleDB)	TimescaleDB hypertable partitioned by time	High-frequency telemetry events; efficient time-range queries; automatic	Retention policy: 30 days; archived to
Storage: Vector DB (Pinecone/Milvus)	High-dimensional face embedding vectors; ANN search	Candidate re-identification across sessions; similarity matching	Embeddings are irreversible; AES-256 at rest
Storage: Object (AWS S3 + SSE-KMS)	Private S3 buckets; SSE-KMS server-side encryption	Session recordings (if consented); audit logs; model artifacts	Pre-signed URL access only. Immut

11.4 Five-Channel Behavioral Consistency Fusion

CHANNEL	MODEL	FEATURES	ANOMALY THRESHOLD	WEIGHT IN FUSION
Visual-Face (Gaze)	MediaPipe Iris + L2CS-Net + Kalman filter	Gaze vector, head pose, fixation duration, saccade frequency	KS-test p-value < 0.05 vs. personal baseline	25% — high specificity
Visual-Body (Gesture)	BlazePose 33kp + MediaPipe Hands 21kp	Wrist velocity, hand-to-face distance, self-touch frequency, bounding box area	Self-touch > 3 SD above baseline; hand beyond 80% frame boundary (30 frames)	20% — high precision
Audio (Forensics)	GMM-UBM speaker verification + MFCC analysis	Voice consistency, ambient noise, response latency, pitch	Voice similarity < 0.6	15% — high recall
Keystroke (Dynamics)	DOM session replay + typing pattern	WPM, dwell time, flight time, correction rate, paste events	WPM > 2.5 SD above baseline; paste detected	20% — high precision
Content (NLP / Semantic)	BERT/RoBERTa vocabulary analysis	Vocabulary sophistication, response coherence, formality	Vocabulary > 2 SD above baseline; coherence drop	20% — high precision

FUSION RULE: Integrity event raised ONLY when N >= 3 channels simultaneously exceed their individual thresholds within the same 10-second analysis window. Severity: N=2 (LOW, logged only), N=3 (MEDIUM, yellow flag), N=4 (HIGH, amber flag + human review), N=5 (CRITICAL, red flag + session pause + mandatory human review before continuation).

12. LEGAL AND ETHICAL LANDSCAPE — 8 GLOBAL FRAMEWORKS

FRAMEWORK	JURISDICTION	KEY REQUIREMENT	MAX PENALTY	2026 STATUS	PROCTORING IMPACT
EU AI Act (Reg. 2024/1689)	EU 27 members + extraterritorial	FRIA, human oversight, candidate notice, bias testing, logging. PROHIBITS emotion recognition in	35M EUR or 7% global turnover	High-risk rules: Aug 2026	Remote AI monitoring (Annex III(4))
NYC Local Law 144 (Int. 1894-A)	New York City, USA	Annual independent bias audit, 10-day written notice to candidates, summary published, alternative	$500–1,500/day	Active since July 2023	First specific AI hiring law. Establishes
Colorado AI Act (SB 24-205)	Colorado, USA	Reasonable care to avoid algorithmic discrimination in high-risk employment decisions.	$20,000/violation	Active since Feb 2026	First comprehensive US state AI law. C
Illinois HB 3773	Illinois, USA	AI discrimination coverage through expanded Human Rights Act. Enforcement via IHRA.	Per violation	Particularly relevant given BIPA (Biome	
UK ICO + DPDI	United Kingdom	AI auditing, bias testing requirements, UK GDPR compliance.	GBP 17.5M / 4%	Equality Act 2010 applies to AI outputs.	Post-Brexit framework mirrors GDPR bu
GDPR (EU 2016/679)	EU + extraterritorial	Lawful basis for biometric data (Art. 9), right to erasure, data minimization, 90-day retention, device processing, consent	20M EUR or 4% global turnover	Active	Requires explicit consent
Canada AIDA (Bill C-27)	Canada (federal)	Transparency, accountability, impact assessments for high-impact systems.	Up to CAD 25M	Proposed	High-impact AI systems include automa
Singapore Model AI (Version 2)	Singapore	Voluntary governance framework: human oversight, explainability, fairness testing.	Voluntary (advisory)	Active since 2020	Sets regional precedent. Major APAC h

HireVue Precedent (2019-2021): EPIC FTC complaint alleged HireVue used secret algorithms to assess 'cognitive ability,' 'psychological traits,' and 'social aptitudes' from facial recognition. Public pressure led HireVue to remove ALL facial analysis features in early 2021 — before the EU AI Act codified this prohibition into law. This case established the industry standard: behavioral proctoring YES, emotion/personality scoring from biometrics NO.

13. WORLD-CLASS MASTER PROMPT — FOR AI RESEARCH AGENTS

The following is the definitive research and development prompt for AI systems tasked with advancing the state of the art in AI interview room technology. It is structured to produce the highest-quality, most comprehensive output possible from any frontier AI model.

MASTER PROMPT — AEGIS INTERVIEW INTEGRITY SYSTEM

```
// ROLE + CONTEXT
You are Dr. AEGIS — the world's foremost expert in AI-powered interview integrity systems. You hold combined expertise in: computer vision (MediaPipe, BlazePose, CAST architectures), behavioral biometrics (gaze estimation, gesture recognition, keystroke dynamics), security engineering (edge-cloud hybrid systems, WebAssembly, WebGPU), applied statistics (Kolmogorov-Smirnov tests, Kalman filtering), deepfake forensics (CNN-Transformer architectures, rPPG, PLR), and biometric privacy law (EU AI Act, BIPA, GDPR, NYC Local Law 144).

// KNOWLEDGE FOUNDATION
You have complete knowledge of:
- Commercial platforms: HireVue (30M+ interviews, Fortune 100 dominant), Talview (7-layer framework, 95% detection), Mercer Mettl, Goodfit, CodeSignal, Fabric (20+ forensic signals, 85% detection), Pymetrics/Harver
- Cheating tools: Cluely/Interview Coder ($5.3M raised, zero detections claimed), FinalRound AI, Natively, ClearRound.ai (Rs.199/week India)
- Attack vectors: compositor-layer overlays, KVM proxy networks, deepfake Tier 1/2/3, virtual machines, Bluetooth earpieces, AR glasses, PLR-defeating liveness bypasses
- Detection systems: MediaPipe FaceMesh (468 landmarks), BlazePose (33 keypoints, 1.3-3.5M params, 31 FPS), MediaPipe Iris + Kalman (97.5% accuracy), L2CS-Net gaze, CAST deepfake (99.49% AUC FF++, 74.92% Celeb-DF), LipFD/SAVe lip-sync (95.3%), rPPG heartbeat, PLR pupil reflex
- The 5 unsolved problems: whitespace fixation detection, per-candidate KS-test gaze baseline, combined PLR+rPPG single check, cross-dataset deepfake generalization, five-channel behavioral consistency fusion

// TASK SPECIFICATION
For every request in this session, you will:
1. FIRST identify which of the 5 unsolved problems the request touches, if any
2. PROVIDE technical implementations at production code level, not pseudocode
3. CITE specific metrics (accuracy, latency, FPS, AUC, TPR, FPR, EER) for every claim
4. IDENTIFY the residual attack surface after each countermeasure
5. SPECIFY the false-positive risk and mitigation for every detection mechanism
6. DISTINGUISH between what commercial systems do vs. what the research frontier requires
7. NEVER recommend a single-modal defense — always specify the fusion requirement
8. FLAG legal compliance implications (EU AI Act Article 5(1)(f), BIPA, GDPR) for every biometric approach

// OUTPUT FORMAT RULES
Structure every response as:
[TECHNICAL ANALYSIS] — The mechanism, the math, the implementation
[RESIDUAL ATTACK SURFACE] — What this does NOT catch and why
[FUSION REQUIREMENT] — Minimum N channels required for this flag
[FALSE POSITIVE RISK] — Rate and mitigation
```

[NOVEL CONTRIBUTION] — Whether this advances the state of art beyond commercial systems

[LEGAL COMPLIANCE] — Applicable regulatory constraints

[CODE SKELETON] — Python/JS implementation skeleton when applicable

// SPECIFIC RESEARCH DIRECTIVES

When researching gaze detection:

- Default to KS-test per-candidate baseline, NOT universal angle thresholds
- Always include whitespace fixation zone mapping as a parallel signal
- Specify Kalman filter state dimensions and noise parameters

When researching liveness detection:

- ALWAYS combine PLR + rPPG in a single check — never one alone
- Specify the screen flash timing sequence for PLR (randomize frequency 0.5-2.0 Hz)
- Specify the RGB channel weights for rPPG (typical: 0.3R + 0.59G + 0.11B for luminance)

When researching gesture detection:

- Always use BlazePose (33kp) + MediaPipe Hands (21kp) together
- Specify the kinematic thresholds: wrist beyond 80% frame = off-camera; ear distance < 80px for 60 frames = earpiece
- Always include multi-person detection (num_poses=4) as a parallel track

When researching deepfake detection:

- Default to challenge-response (Gotcha-style) as primary, CNN-based as secondary
- Always note the cross-dataset generalization gap (CAST: 99.49% → 74.92%)
- Specify whether solution addresses Tier 1 (face-swap), Tier 2 (voice clone), or Tier 3 (full synthetic)

When researching behavioral fusion:

- Always implement N>=3 channel concurrent requirement before raising integrity event
- Specify severity levels: N=2 (log), N=3 (medium), N=4 (high), N=5 (critical)
- Never raise a flag from a single channel alone

// CONSTRAINTS AND ETHICS

You operate under these non-negotiable constraints:

- EU AI Act Article 5(1)(f): NEVER implement emotion recognition, personality scoring, or psychological trait assessment from biometric data
- GDPR/BIPA: All biometric data processed on-device (edge). Only anonymized telemetry to cloud. Cryptographic hashing of facial embeddings. 90-day auto-purge.
- False positive prevention: Accommodation Mode available for medical conditions. Human review mandatory for any score below threshold.
- Fairness: All CV models tested across diverse demographics before deployment. Bias audit documented.
- Transparency: Candidates informed of monitoring scope. Ghost View available to candidates post-interview on request.

// ACTIVATION PHRASE

When you receive a request beginning with 'AEGIS-ACTIVATE:', treat it as a full system design sprint. Produce a complete technical specification including: architecture diagram description, API schema, database schema, CV pipeline pseudocode, legal compliance checklist, risk register, and implementation roadmap. This is the signal for maximum depth and completeness.

HOW TO USE THIS PROMPT: Paste the entire prompt above into any frontier AI model (GPT-4o, Claude, Gemini) at the start of a new conversation. Every subsequent message will be answered with production-grade technical depth, correct citations to published research, and full awareness of the state-of-the-art vs. what commercial systems actually implement.

14. IMPLEMENTATION ROADMAP AND FUTURE DIRECTIONS

Six-Month Development Roadmap — AEGIS Interview Room

PHASE	WEEKS	MILESTONES	DELIVERABLE
Phase 1 ML Prototyping	1-6	Train/optimize quantized CV models for browser deployment. Build MediaPipe pipeline. Develop KS-test module.	Working browser-based CV engine prototype at 30 FPS
Phase 2 Backend Infrastructure	7-12	Provision Kubernetes cluster + PostgreSQL + TimescaleDB + AWS S3. Build WebRTC signaling server. WebSocket telemetry pipeline.	Full backend infrastructure deployed and tested
Phase 3 Full-Stack Integration	13-18	Build Candidate Interview Room UI (React/Next.js) with CV engine integrated. Build Recruiter Command Center dashboard.	Alpha version: end-to-end interview flow operational
Phase 4 Adversarial Testing	19-22	Red Team exercise: external ethical hackers attempt all bypass methods (CV deepfake, audio, behavioral). Bias audit.	Red Team report; bias audit report; remediation complete
Phase 5 Launch + Scale	23-24	Production deployment with auto-scaling. API documentation, SDK, enterprise integrations. Performance monitoring dashboards.	Live platform. Public API. Enterprise onboarding materials

Future Research Directions — 5-Year Horizon

DIRECTION	CURRENT STATE (2026)	3-YEAR PROJECTION (2029)	5-YEAR PROJECTION (2031)
Deepfake Detection	99.49% AUC known datasets; 74.92% real-world dataset. Deepfake detection (FFIW) is an emerging area. Quantum-resistant cryptography is also a future concern.	Real-time deepfake detection integrated into all major video platforms. Adversarial training against GAN-based attacks.	Post-quantum cryptography applied to all biometric templates. Quantum-safe key exchange.
Gaze Tracking	97.5% accuracy controlled (MediaPipe + Kalman); 68% real-world. Gaze-based cognitive load estimation.	3D gaze tracking with depth estimation. Gaze-based reading detection. Integration with AR/VR headsets.	Neural gaze models trained on millions of sessions. Predictive gaze for UI optimization.
Behavioral Biometrics	99.85% accuracy session-long authentication. Keystroke dynamics, mouse trajectory, and touch gesture analysis.	Cross-device behavioral profiles. Continuous authentication without any explicit challenge.	Behavioral biometrics as a standalone identity layer. Integration with decentralized identity systems.
Privacy Technology	Edge processing emerging. Federated learning for model training. Differential privacy for aggregate analytics.	Full on-device inference. Homomorphic encryption for server-side processing. Zero-knowledge proofs for age verification.	Fully private AI: no raw biometric data ever leaves the device. Verifiable computation for audit trails.
Legal / Regulatory	8 active frameworks. EU AI Act high-risk classification. Global convergence toward mandatory AI auditing.	20+ jurisdictions. Mandatory third-party audits for all high-risk AI systems. Standardized bias benchmarks.	Unified global AI governance framework. Real-time regulatory compliance monitoring. Automated audit reporting.

Resource Requirements for AEGIS Development

ROLE	COUNT	PRIMARY RESPONSIBILITY
Project Lead / Architect	1	System architecture, technical direction, stakeholder management
CV / ML Engineers (Edge Inference)	3	MediaPipe pipeline, ONNX optimization, liveness algorithms, KS-test module
Backend Engineers (WebRTC + Data)	2	WebSocket telemetry, Kafka pipeline, FastAPI microservices, TimescaleDB
Frontend Engineers (React + WebGL)	2	Candidate interview UI, recruiter dashboard, Ghost View, WebGPU integration
UX / UI Designer	1	Minimalist cognitive-load-reduction design; WCAG 2.1 AA compliance
Security / Compliance Specialist	0.5	GDPR/BIPA/EU AI Act compliance; IAM; encryption; audit logging
QA / Red Team	2	Adversarial testing; bypass simulation; bias audit coordination

AEGIS RESEARCH INSTITUTE AI Interview Room: Anatomy of Fraud & Defense Deep Technical Intelligence Report | v3.0 FINAL | June 2026

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